

# Geongyu Lee 이건규 · イ · ゴンギュ

## AI Researcher · Computational Pathology × Multi-Omics

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Five years building deep learning for computational pathology and drug discovery. First-author work on H&E-based breast-cancer recurrence (Scientific Reports 2025), a contribution to large-scale pathology foundation models (G2L, AAAI 2026 oral), and MFDS-track model validation; now extending histopathology AI into proteomics and multi-omics.

### SELECTED PUBLICATIONS

Google Scholar · 55 citations · h-index 5 · i10-index 2 · as of Sep 2026

- 1 Recurrence risk prediction in early-stage breast cancer from H&E whole-slide images.** Scientific Reports, 2025. First author · doi.org/10.1038/s41598-025-16679-x
- 2 G2L: From Giga-scale to Cancer-specific Pathology Foundation Models via Knowledge Distillation.** AAAI 2026 Workshop (W3PHIAI), Oral. arXiv:2510.11176
- 3 KPIs 2024 Challenge: Advancing glomerular segmentation from patch- to slide-level.** Medical Image Analysis 114, 104234, 2026. 2nd place, whole-slide track · doi.org/10.1016/j.media.2026.104234
- 4 AI-driven digital pathology in urological cancers (Review).** Prostate International, 2025. Co-first author
- 5 MurSS: Multi-Resolution Selective Segmentation for breast cancer.** Bioengineering, 2024. MDPI
- 6 Spatial proteomics guided by H&E-based AI reveals recurrence-risk niches in triple-negative breast cancer.** Preprint, 2026. arXiv:2608.03145
- 7 MoSPR: Histology-to-gene expression prediction with morpho-spatial macrostates and low-rank molecular programs.** Preprint, 2026. Co-first author · arXiv:2609.34280 · code released
- 8 Multi-section WSI analysis for biochemical recurrence in prostate cancer.** Preprint, 2026. arXiv:2603.20273
- 9 Supervised contrastive embedding for medical image segmentation.** IEEE Access, 2021.

### EXPERIENCE

#### OMIXAI (fmr. RadiSen)

Feb 2025 – Present

AI Researcher · Seoul

- Lead multi-omics foundation-model R&D integrating proteomics with H&E pathology for drug-response prediction; co-authored the G2L pathology foundation model.
- Co-authored spatial-proteomics work using H&E-based AI risk scores to pinpoint recurrence-risk niches in triple-negative breast cancer (preprint, 2026).
- Built a proteomics drug-response model (Leave-Drug-Out Pearson  $\geq 0.65$ ) and a canine veterinary-oncology drug-recommendation algorithm (top-k  $\geq 70\%$ ).
- Advanced ADMET prediction and self-supervised proteomic representation learning with LoRA / PEFT.
- Co-led the Virtual Cell Challenge: predicting CRISPR-knockdown response in pluripotent stem cells.

#### Deep Bio

Mar 2021 – Jan 2025

AI Researcher · Seoul

- First-authored the multi-center breast-cancer recurrence study (Scientific Reports 2025); built a scalable WSI pipeline handling 500+ slides.
- Developed lymph-node metastasis detection and the KPIs 2024 glomeruli segmentation model (2nd place, whole-slide track, MICCAI); challenge results published in Medical Image Analysis (2026).
- Led prostate metastasis & recurrence-risk projects; presented at AACR (2022, 2023) and USCAP (2022).
- Contributed to MFDS regulatory documentation and validation, plus internal server-automation tooling (Docker).

#### Nuricon

2021

Intern · Pangyo

- Built a parking-lot fire-detection AI system.

### SELECTED PROJECTS

#### Proteomics drug-response prediction · OMIXAI

Self-supervised proteomic representation with LoRA/PEFT and cell-line IC50 prediction; leave-drug-out validation (Pearson  $\geq 0.65$ ).

**Veterinary oncology CDSS & Virtual Cell** · OMIXAI

Canine drug-recommendation (top-k ≥ 70%) and CRISPR-knockdown response prediction in pluripotent stem cells.

**FlyGate: drug-discovery → pharmacovigilance agent** · NVIDIA Agentic AI Hackathon 2026

Side project (team of 5): built FlyDiscovery on BioNeMo NIM (OpenFold3 1.0 Å, Boltz-2 p 0.767) and its UI.

**Oncotype DX recurrence prediction** · Deep Bio

H&E-only breast-cancer prognosis; confidence-aware selection + majority voting into GHI-RS risk groups (high-risk acc 91.2%).

**Brain-hemorrhage detection on CT** · SK / Ajou Univ. Hospital

2D & 3D hemorrhage segmentation, slice-level classification, domain-generalization validation with class-activation-map auditing.

**EDUCATION**

**M.S. Data Science · Seoul Nat'l University of Science and Technology (SeoulTech)**

2019 – 2021

Advisor: Prof. Sangheum Hwang · Thesis: contrastive loss for segmentation under uncertain medical-image labels.

**SKILLS**

|                              |                                                                                    |
|------------------------------|------------------------------------------------------------------------------------|
| <b>Deep Learning</b>         | PyTorch, HuggingFace, PEFT / LoRA, Accelerate, DDP / FSDP                          |
| <b>Pathology</b>             | OpenSlide, QuPath, CLAM / weakly-supervised MIL, UNI · CONCH · Virchow, WSI tiling |
| <b>Bio / Omics</b>           | scanpy · AnnData, single-cell perturbation, proteomic representation, ADMET        |
| <b>MLOps &amp; Languages</b> | Python, R, SQL, Bash, Docker, Slurm, W&B, MLflow, FastAPI, Git                     |

**AWARDS & FUNDING**

**KPIs Challenge 2024 · Whole-Slide Track**

2nd place · Glomeruli segmentation, held at MICCAI 2024 · results published in Medical Image Analysis 2026

**Pan-Sarcoma Proteogenomic Profiling for Precision Oncology**

Korea–US–Japan · Kyung Hee Univ. Medical Center · 2025–2028 · KRW 400M · Participating Researcher

**Virtual-Cell CDSS for Veterinary Oncology**

IPET / Ministry of Agriculture · 2026–2030 (awarded) · KRW 3B+ · Participating Researcher

**General-Purpose AI for Cancer Pathology Diagnosis**

Lead: Deep Bio · National R&D · 2021–2025 · KRW 2.375B · Participating Researcher

**CONFERENCE PRESENTATIONS**

BIOINFO/GIW ISCB-Asia 2026 · Accepted poster · First & presenting author · Predictability is not substitutability: a cost-of-substitution framework for H&E-based molecular prediction across 5 cancers

BIOINFO/GIW ISCB-Asia 2026 · Accepted oral · Co-author · A reliability map for per-gene multiome RNA velocity parameters in single-cell kinetics

AACR Annual Meeting 2023 · Poster · First author · Predicting protein receptor status from H&E-stained images in breast cancer

AACR Annual Meeting 2022 · Poster · First author · Recurrence risk prediction based on automatic histologic analysis of breast cancer using whole slide images

AACR Annual Meeting 2022 · Poster · Co-author · A deep learning based pancreatic adenocarcinoma survival prediction model applicable to adenocarcinoma of other organs

USCAP Annual Meeting 2022 · Poster · First author · Automatic histological grading of breast cancer resection tissue

USCAP Annual Meeting 2022 · Poster · Co-author · Breast cancer survival analysis through the extracted feature from the prostate diagnosis model

KIIE Fall Conference 2020 · Oral · First author · Utilizing a contrastive loss to improve segmentation model performance

**COMMUNITY & SERVICE**

Research member (Runner), Pseudo Lab season 12: AutoBioX, AI Agents for End-to-End Bio Research · 2026 · 16 weeks · two outputs accepted at GIW ISCB-Asia 2026, then submitted to ML4H 2026 Findings

Reviewer, ML4H 2026 (Machine Learning for Health Symposium) · 2026

AI Career & Project Mentor, Codeit · 2025 – present · part-time · 13 mentees 1:1, 8+ project teams across 2+ cohorts